Encoded does not mean used: Causal separation of subject and attractor number in a Transformer

A language model predicting the verb in (1) has access to two number values: the hierarchical subject is singular, while the linearly nearest noun is plural. Behavioral work shows that Transformers are sensitive to this distinction, but mechanistic studies face a basic inference problem. A feature may be decodable without being used, and a head may display syntactic attention without controlling the output (Elazar et al. 2021; Timkey & Linzen 2026). We use agreement attraction to separate these possibilities. In a controlled DistilGPT-2 pilot, nearest-noun number becomes perfectly decodable, yet erasing a single subject-number direction—not the attractor direction or matched random directions—selectively damages agreement. The result supplies causal evidence that hierarchical and linear number cues coexist in the residual stream but have different functional status.

(1) *The key near the cabinets {is/are}...*

Question and predictions. Causal analyses have localized agreement information to neurons, heads, and residual-stream directions (Finlayson et al. 2021; Ferrando & Costa-jussà 2024). Separately, GPT-2 exhibits attraction effects compatible with similarity-based interference (Ryu & Lewis 2021). What remains unclear in a small, inspectable model is whether a highly accessible attractor representation is also the representation that drives the inflection. A purely diagnostic account predicts that both numbers can be decoded but makes no prediction about use. A nearest-noun mechanism predicts that removing attractor information should be at least as consequential as removing subject information, especially when the two conflict. A hierarchical mechanism instead predicts a selective effect of subject-number erasure on incongruent items.

Materials and behavioral result. We generated 80 prefixes from ten lexical sets, crossing subject number (singular/plural), attractor number (singular/plural), and structure. In the PP condition, the attractor was inside a subject-modifying PP as in (1); in the relative-clause condition it was the embedded subject, as in *The key that the cabinets can describe*. All target forms, *is* and *are*, are single tokens. We measured their next-token logit contrast in DistilGPT-2 (81.9M parameters). The model selected the grammatical form on 92.5% of items (74/80). Accuracy was 97.5% for number-congruent items but 87.5% for incongruent items; it was 100% in PPs and 85% in relative clauses. Thus the model is accurate enough for an intervention, while retaining the expected attraction signature.

Diagnostic analysis. At the final prefix position, we extracted the embedding state and the residual stream after each of six Transformer blocks. For every state, separate logistic probes predicted subject and attractor number. Five-fold cross-validation held out both structures and all four number conditions for two complete lexical sets per fold. Table 1 shows that subject number rises from chance at the embedding state to 93.75% after block 5. Strikingly, attractor number is easier: it reaches 100% after blocks 3 and 4. Decodability alone would therefore suggest no privileged status for hierarchical number.

Causal intervention. For each post-block state, we fit subject- and attractor-number probes on the balanced stimulus set, centered each weight vector, and separately projected the final-position residual stream away from each one-dimensional direction. We then reran the unchanged remainder of the model. As a dimensionality-matched control, we repeated the procedure with ten isotropic random directions. Subject erasure has almost no effect in blocks 1–3, then produces increasing correct-margin losses of 0.382, 0.817, and 1.739 logits after blocks 4–6. The final effect is stable under a paired 10,000-sample bootstrap, 95% CI [1.454, 2.017]. Accuracy falls from 92.5% to 76.25%; on the diagnostic incongruent items it falls from 87.5% to 62.5%. Attractor erasure leaves both accuracies unchanged and changes the final margin by only 0.005 logits; random erasure likewise leaves accuracy at 92.5% and changes the margin by 0.002.

Table 1: Group-cross-validated probe accuracy and causal change in the correct-inflection logit margin. State 0 is the embedding output.

State	Subj. probe	Attr. probe	Subj. erase	Attr. erase	Random
0	50.00	75.00	–	–	–
1	71.25	86.25	−0.000	0.004	0.000
2	58.75	97.50	0.012	−0.006	0.002
3	73.75	100.00	0.030	−0.018	0.001
4	91.25	100.00	0.382	−0.014	0.000
5	93.75	97.50	0.817	0.003	0.002
6	92.50	88.75	1.739	0.005	0.002

Analysis and contribution. The probe results establish coexistence: the representation preserves both the structurally licensed controller and a salient linear distractor. The intervention establishes an asymmetry in use. The model's agreement preference depends on a late-emerging subject-number direction even though attractor number is more linearly accessible at intermediate states. This dissociation clarifies what agreement *knowledge* amounts to mechanistically: robust encoding of a cue is not sufficient evidence that the cue participates in the computation.

The late locus also connects two earlier findings. Finlayson et al. (2021) report structure-dependent agreement mechanisms, while Ferrando and Costa-jussà (2024) identify a subject-number direction written into the final residual stream. Our result shows why a competing noun can remain decodable without controlling English copular inflection: hierarchical selection can be implemented by a targeted late readout rather than deletion of the distractor. The proposal is deliberately limited to one model and 80 items. Follow-up work should cross-fit the intervention direction, test additional agreement paradigms and model seeds, and compare direct direction erasure with more targeted methods such as mean projection or LEACE (Dobrzeniecka et al. 2025). Even at pilot scale, however, the encoding/use dissociation is large, controlled, and linguistically interpretable.

References

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